

Skill13.14.15.16

13.To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

1. INPUT:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        int fd = open("input.txt", O_RDONLY);

        if (fd < 0) {
            perror("open");
            exit(1);
        }

        dup2(fd, STDIN_FILENO);
        close(fd);

        execlp("wc", "wc", "-l", NULL);

        perror("execlp");
        exit(1);
    }

    wait(NULL);

    return 0;
}
```

```
sharanaya@DESKTOP-TPK9V3Q nano skill13.c
sharanaya@DESKTOP-TPK9V3Q gcc skill13.c -o skill13
sharanaya@DESKTOP-TPK9V3Q ./skill13
open: No such file or directory
```

14. To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

2. INPUT:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        int out = open("output.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

        int err = open("error.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

        if (out < 0 || err < 0) {
            perror("open");
            exit(1);
        }

        dup2(out, STDOUT_FILENO);
        dup2(err, STDERR_FILENO);

        close(out);
        close(err);

        execlp("ls", "ls", "nonexistent_file", NULL);

        perror("ls");
        exit(1);
    }

    wait(NULL);
    wait(NULL);

    printf("Output written to output.txt\n");
    printf("Error written to error.txt\n");

    return 0;
}
```

output:

```
sharanaya@DESKTOP-TPK9V3Q nano skill14.c
sharanaya@DESKTOP-TPK9V3Q gcc skill14.c -o skill14
sharanaya@DESKTOP-TPK9V3Q ./skill14
Output written to output.txt
Error written to error.txt
sharanaya@DESKTOP-TPK9V3Q |
```

15. To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs.

To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios.

3. INPUT:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int pipefd[2];

    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    pid_t p1 = fork();

    if (p1 == 0) {
        int fd = open("result.txt",
                     O_WRONLY | O_CREAT | O_TRUNC,
                     0644);

        if (fd < 0) {
            perror("open");
            exit(1);
        }

        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        execlp("ls", "ls", NULL);

        perror("ls");
        exit(1);
    }
}
```

```

pid_t p2 = fork();

if (p2 == 0) {
    int fd = open("result.txt",
                  O_WRONLY | O_CREAT | O_TRUNC,
                  0644);

    if (fd < 0) {
        perror("open");
        exit(1);
    }

    dup2(pipefd[0], STDIN_FILENO);
    dup2(fd, STDOUT_FILENO);

    close(fd);
    close(pipefd[0]);
    close(pipefd[1]);

    execlp("wc", "wc", "-l", NULL);

    perror("wc");
    exit(1);
}

close(pipefd[0]);
close(pipefd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

printf("Result stored in result.txt\n");

return 0;
}

```

OUTPUT:

```

sharanaya@DESKTOP-TPK9V3Q nano skill15.c
sharanaya@DESKTOP-TPK9V3Q gcc skill15.c -o skill15
sharanaya@DESKTOP-TPK9V3Q ./skill15
Result stored in result.txt
sharanaya@DESKTOP-TPK9V3Q |

```

16. To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

INPUT:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background process started. PID = %d\n", getpid());

        sleep(5);

        printf("Background process completed.\n");

        exit(0);
    }

    printf("Parent continues immediately.\n");
    printf("Background PID = %d\n", pid);

    printf("Parent is not waiting immediately.\n");

    sleep(2);

    waitpid(pid, NULL, 0);

    printf("Background job finished.\n");

    return 0;
}
```

output:

```
sharanaya@DESKTOP-TPK9V3Q nano skill16.c
sharanaya@DESKTOP-TPK9V3Q gcc skill16.c -o skill16
sharanaya@DESKTOP-TPK9V3Q ./skill16
Parent continues immediately.
Background PID = 1356
Parent is not waiting immediately.
Background process started. PID = 1356
Background process completed.
Background job finished.
sharanaya@DESKTOP-TPK9V3Q |
```